

Plasmonic Evolution and Arrested Development for Silver Nanoscale Colloids: A Classroom Demonstration

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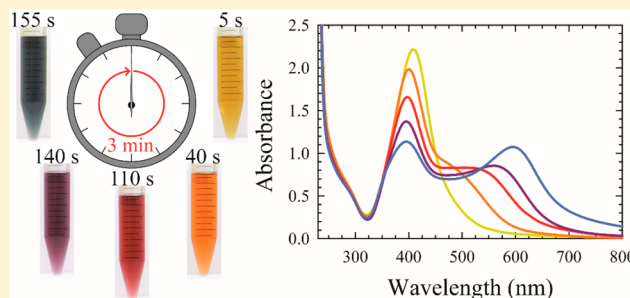
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S Supporting Information

ABSTRACT: A vivid demonstration visualizing the colloidal behavior of silver nanocrystals is described. An aqueous–ethanol silver ion solution becomes yellow upon initial chemical reduction (using sodium borohydride, NaBH_4), which is followed by chromatic evolution to orange, red, violet, and finally blue over the course of approximately 3 min because of the predictable growth and evolution of the nanocrystals. This undergraduate-level classroom demonstration is useful for illustrating the protective effect of a passivating ligand, such as polyvinylpyrrolidone (PVP), by showing the ability of PVP to halt the growth and coalescence of silver colloids at a desired stage, arresting the color transition at a particular hue. This demonstration can also be extended to highlight the bifurcated impact of such passivation on colloid behavior (e.g., improved long-term colloid stability coupled with worsened nanocatalytic performance).

KEYWORDS: First-Year Undergraduate/General, Second-Year Undergraduate, Demonstrations, Inquiry-Based/Discovery Learning, Metals, Colloids



INTRODUCTION

Seasoned lecturers can appreciate the improvement in concept retention provided by visual demonstrations in undergraduate courses.^{1,2} The positive effects of science demonstrations on the engagement, emotional energy, and learning of students are particularly effective with striking visuals, often employing colorful model reactions for this purpose, the iodine clock reaction^{3–5} being the most notorious example. Some additional recent examples of visual classroom demonstrations from this *Journal* include the electrolytic dissolution of a copper anode;⁶ the primary kinetic isotope effect in the Cr(VI) oxidation of 2-propanol- d_8 or methanol- d_4 ;⁷ plasma enhanced chemical vapor deposition patterning of hydrophobic and hydrophilic surfaces and visualization with food dye to yield “aqua-art”;⁸ “hydroglyphics” prepared via selective wetting on hydrophilic and hydrophobic surfaces using masked corona treatment with a modified Tesla coil;⁹ oxidation of tetramethylbenzidine with hydrogen peroxide catalyzed by Fe_3O_4 nanoparticles followed by catalyst retrieval for reuse using an external magnet;¹⁰ and color-changing goldenrod paper (e.g., folded into “flowers”) responsive to pH, sodium hypochlorite, iodine, and vitamin C.¹¹ In this demonstration, in a modest time investment of less than 5 min, we provide a visual nanochemistry experience that vividly illustrates the color evolution (surface plasmon resonance) accompanying nanoscale silver growth, highlighting the effects of colloid aggregation mediated by the surface chemistry (capping ligands) of silver nanocrystals (NCs).

Precious metal NCs have been appreciated since antiquity because of the vibrant colors and tunable photonic properties of their colloids. For example, the Romans famously used gold and silver NCs (AuNCs and AgNCs, respectively) in the manufacturing of red (e.g., the Lycurgus cup) and yellow stained glass, respectively.^{12,13} This distinct visual appeal relies entirely on well-defined plasmonic properties, dictated by NC size,^{14–16} shape,^{16,17} surfactants,^{18,19} and concentration,²⁰ each of which has been rigorously studied over the last half century. As shown by Zielińska et al., silver colloids exhibiting diverse plasmonic features are possible using chemical reduction routes involving various combinations of silver precursor (e.g., nitrate, acetate, or citrate salts), reducing agent (e.g., ascorbic acid, hydrazine, sodium borohydride), and stabilizing agent (e.g., polyvinyl pyrrolidone, polyvinyl alcohol).²¹ Although many publications in this *Journal* address NC systems^{14,22,23} and silver specifically,^{16,17,24–26} these studies promote rapid synthesis and application, with NC aggregation being shown as a preventable outcome rather than a pedagogical focus. We propose that an appropriate visual demonstration involving growth and aggregation of colloidal AgNCs should provide a color transition that lasts for several minutes, allowing for extended observation and discussion of the relevant chemistry. Unfortunately, most AgNC solutions reported in the literature age far too rapidly, and a prolonged classroom demonstration

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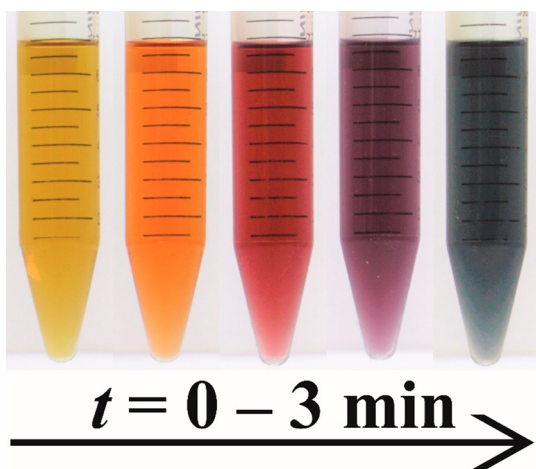


Figure 1. Color progression of the NaBH_4 -stabilized AgNC colloid. The total time for the transition from yellow to blue is approximately 3 min.

with a vibrant and visually appealing colloid has yet to be reported. Here, we provide such a demonstration: a simple reaction between AgNO_3 and NaBH_4 in a water/ethanol mixture to prepare a yellow colloid of AgNCs. This solution slowly undergoes a prominent and striking color change to orange, red, violet, and finally blue over 3 min as the AgNCs aggregate. The 3 min time frame provides ample time for evaluation and discussion of the nanoscale reaction, making this reaction of tremendous academic value as a visual aid for teaching about colloidal NC systems. Further, the color progression can be arrested by addition of polyvinylpyrrolidone (PVP), a common stabilizing ligand,²⁷ at any point during AgNC aggregation. Although the vibrant colors of this demonstration can easily spark interest in a younger audience, we performed this demo in an undergraduate-level Quantitative Chemical Analysis course alongside pertinent discussions interrelating the plasmon resonance bands, sizes, and catalytic activities of noble metal nanoparticles.

■ EXPERIMENTAL OVERVIEW

Experimental Directions

The full experimental procedure can be found in the [Supporting Information](#). In short, 17.20 mL of 200 proof ethanol (EtOH) and 0.80 mL of ultrapure H_2O were combined in a 50 mL polyethylene centrifuge tube. Next, 1.00 mL of 5.0 mM ethanolic AgNO_3 was added, and the centrifuge tube was vortexed for 30 s to ensure thorough homogenization. Finally, 1.00 mL of 25.0 mM ethanolic NaBH_4 was rapidly added under vortex mixing, resulting in a final solvent ratio of 4:96 $\text{H}_2\text{O}/\text{EtOH}$. The reaction solution initially appears vibrant yellow upon synthesis of the AgNCs, after which there is a visually pleasing color transition to orange, red, violet, and finally blue over 3 min, as shown in [Figure 1](#). The blue color becomes progressively fainter over 24 h as the AgNCs aggregate, ultimately resulting in the formation of black aggregates that precipitate from the solution. Further, this color progression can be arrested at any point in the reaction by rapid addition of 1.00 mL of 5.0 mM ethanolic polyvinylpyrrolidone (PVP, 40 000 Da) while stirring (see [Scheme 1](#)). As a visual demonstration, it may be useful to prepare several colored solutions of PVP-stabilized colloidal AgNCs in advance if more time is needed for discussion. Notably, the reaction solutions should be prepared just prior to use, as NaBH_4 degrades over time. All ethanolic solutions should be stored in sealed containers to minimize evaporation (the NaBH_4 solution should be vented periodically to prevent hydrogen gas buildup).

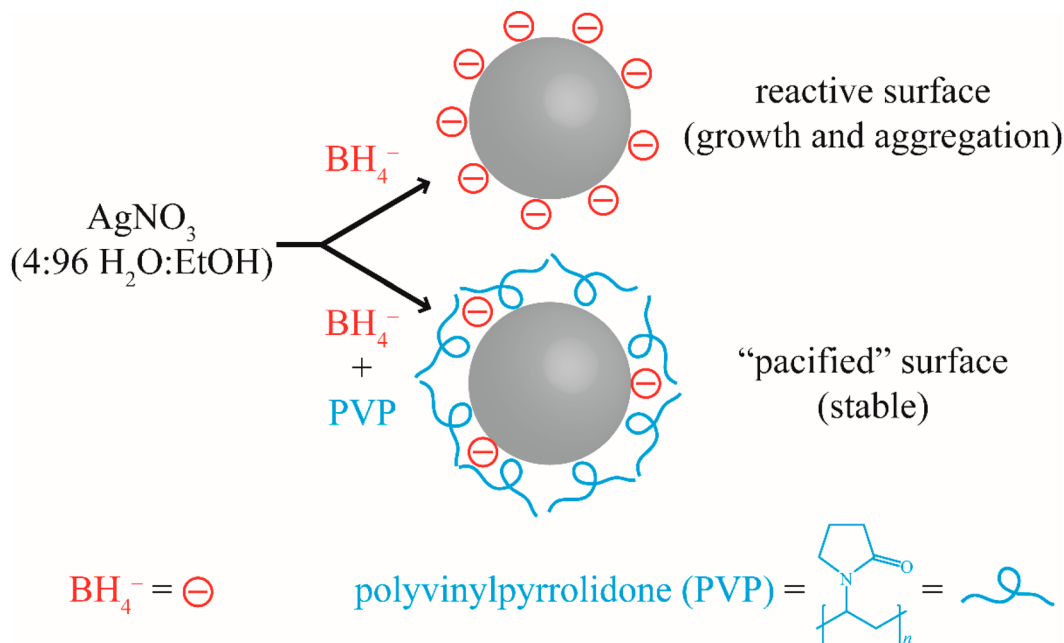
Video Display

If desired, the sequence can be projected with a document camera. In this case, use of a stir plate and magnetic stir bar is proposed to ensure reaction homogeneity. Sample videos can be viewed in the [Supporting Information](#).

■ HAZARDS

Personal protective equipment such as goggles, gloves, and an apron or lab coat should always be worn while preparing and

Scheme 1. Reduction of Ag^+ with BH_4^- To Produce Colloidal AgNCs in the Absence and Presence of PVP as a Steric Stabilizing Ligand



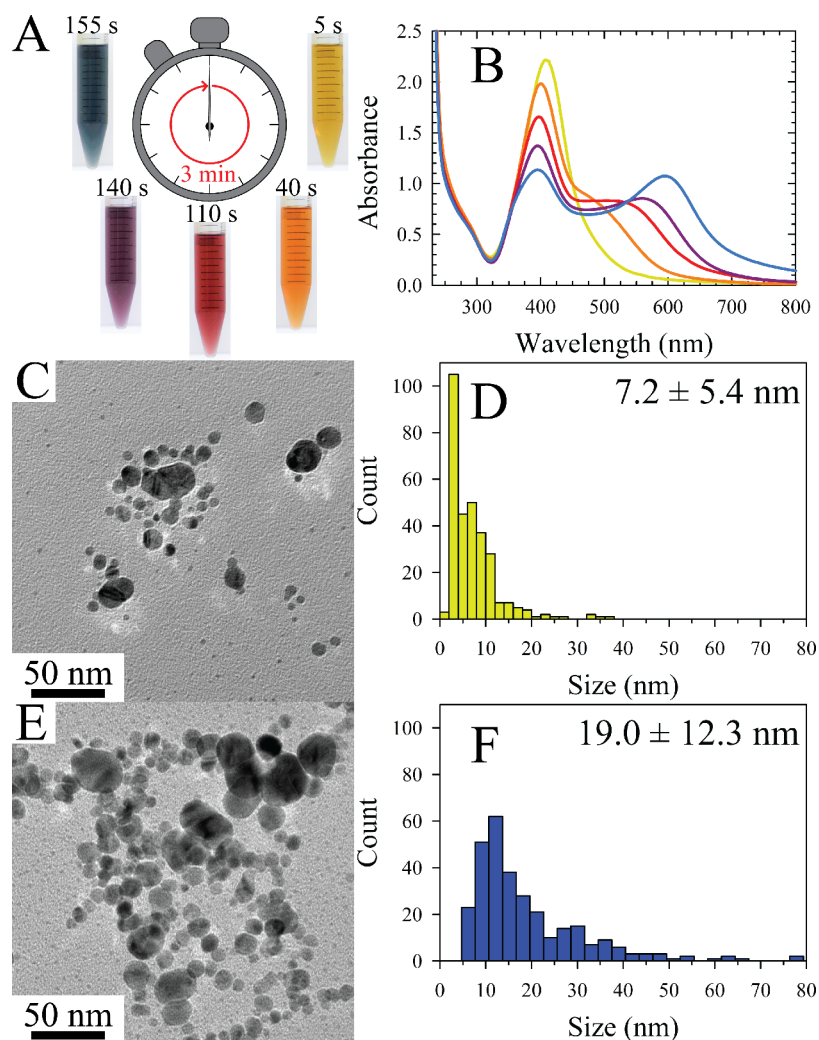


Figure 2. (A) AgNC solution proceeding through five visually distinct colors from yellow to blue over the course of 3 min. (B) UV–vis spectra for each colored solution clearly showing the changes in the localized surface plasmon resonance (LSPR) bands, with the peak at 400 nm blue-shifting and decreasing in intensity as a second peak emerges and intensifies near 550–600 nm. (C,E) TEM images of the AgNCs present in the yellow and blue solutions, respectively, and (D,F) corresponding AgNC size histograms (≥ 300 AgNCs counted for each sample). We note that each TEM image also shows a large population of small AgNCs (below 5 nm) that greatly outnumber the larger particles. Because of their abundance and negligible effect on the LSPR, these ultrasmall AgNCs were not enumerated when generating size distribution histograms.

performing this demonstration. The instructor must ensure all vessels are clearly labeled to avoid confusing reactant solutions. Silver nitrate can act as an oxidizing agent and poses a health risk, especially if ingested. Sodium borohydride is a strong reducing agent and can be corrosive. Further, NaBH_4 solutions naturally decompose to produce hydrogen gas, which will need to be vented. Ethanol fumes are highly combustible, so this visual is best preformed in a well-ventilated area. Polyvinylpyrrolidone is relatively harmless; however, combustion can lead to the development of asphyxiating gases. Because of the flammable nature of the reactants, a fire extinguisher should be nearby at all times. Silver nanocrystals, despite their growing commercialization (i.e., because of their excellent antimicrobial activity, roughly one-third of nanomaterial-containing consumer products contain nanosilver),²⁸ should be treated as toxic and disposed of in a clearly labeled waste container.

RESULTS AND DISCUSSION

The spectral properties of the AgNC colloid were monitored using UV–vis spectroscopy as the solution color transitioned

from yellow to blue. Specifically, the primary peak of the localized surface plasmon resonance (LSPR) band was observed to decrease in intensity and shift from about 410 to 400 nm, whereas a second peak evolved at around 550 nm and shifted to 600 nm as the colloid turned blue (Figure 2B). Although the peak at 400–410 nm is commonly observed for colloidal AgNCs, the secondary peak is likely a product of aggregation and is the proposed cause of the observed color changes of the colloid, although anisotropic growth (e.g., prisms) may contribute in addition to colloid ripening. The spectral shifting halts after 3 min, and the LSPR decreases in intensity over several hours as black aggregates precipitate from solution (Figure S1). The color development was arrested at yellow, orange, red, purple, and blue colloids by the rapid addition of PVP after aging solutions for 5, 40, 110, 140, and 155 s, respectively. The aggregation progress for this reaction was discerned by measuring the average AgNC sizes from these PVP-stabilized colloids using transmission electron microscopy (TEM) imaging. Figure 2 presents the images and respective histograms of AgNCs recovered from the yellow (7.2 ± 5.4 nm)

and blue (19.0 ± 12.3 nm) colloids, with images and histograms of the orange (12.3 ± 7.1 nm) and purple (13.2 ± 7.2 nm) colloids presented in Figure S2.

It was determined that any solvent ratio that contained more than 10% (v/v) water yielded a yellow colloid of AgNCs that aggregated over the next 3 days without demonstrating a vibrant visual color progression. Using pure EtOH resulted in a similar, albeit faster (~ 45 s), color progression to what is shown in Figure 2. We attribute this sensitivity regarding solvent ratio to the rapid decomposition of BH_4^- to hydrogen gas and borohydroxides in the presence of water, a process which is slowed in EtOH. Borohydride is often credited as the stabilizing ligand in aqueous solutions of NaBH_4 –AgNCs,²⁹ and its decomposition leads to aggregation. On the other hand, we propose that the negatively charged borohydride surface requires a hydration layer to slow aggregation of the AgNCs, such that the lower water content samples display rapid color changes. The 4% (v/v) water solvent system used herein provides a vibrant color transition over 3 min, an appropriate amount of time for discussion of the AgNCs and the relevant nanochemistry.

The colloid of AgNCs was aged for 1 h (blue solution), and the catalytic potential of these nanocatalysts was assessed using the model reduction of 4-nitrophenol to 4-aminophenol (Figure 3).

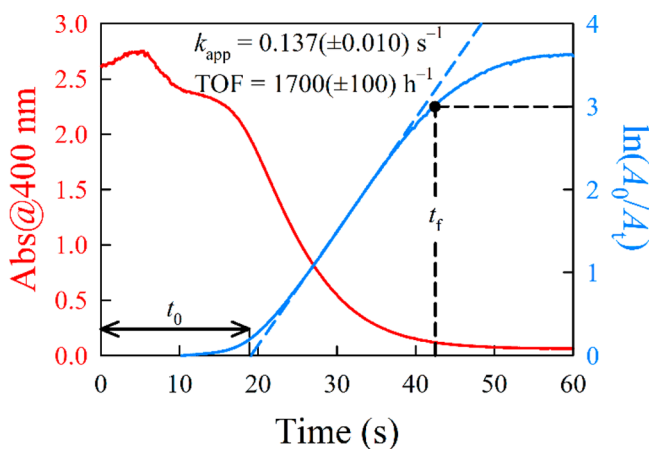


Figure 3. Time-resolved absorbance (left axis) and pseudo-first-order rate plot (right axis) for the catalyzed reduction of 4-nitrophenol to 4-aminophenol using NaBH_4 and colloidal NaBH_4 –AgNCs (aged for 1 h, blue solution). The designated time points of t_0 and t_f represent the induction time (during which dissolved O_2 gas is eliminated in a back reaction with 4-aminophenol³⁶) and the final reaction time (which occurs at $\ln(A_0/A_t) = 3$, i.e., 95% reaction completion), respectively.

Details for the reduction procedure can be found in the Supporting Information. The reaction was monitored spectroscopically following degradation of the 4-nitrophenolate intermediate ($\lambda_{\text{max}} = 400$ nm). The pseudo-first-order rate, k_{app} , is derived from the slope of the linear plot of $\ln(A_0/A_t)$ versus time, where A_0 and A_t are the initial and time-dependent absorbances at 400 nm, respectively. Here, the turnover frequency (TOF) is calculated (as we^{30,31} and others³² have done previously) by dividing the number of moles of converted substrate by the product of moles of catalyst and reaction time (see eq 1 in the Supporting Information). Although some researchers have argued the merits of employing surface area in the TOF calculation,^{33,34} there is currently no universally followed metric for TOF calculation in nanocatalysis.³⁵ Moreover, our approach has pragmatic value in that (i) one can

compare the TOF directly on the basis of total metal within a system (i.e., the economics of noble metal usage are properly accounted for); (ii) it does not require (often inaccessible or tedious) knowledge of the distribution of surface atom types (e.g., faces, edges, corners, and adatoms) nor their precise passivation by surface chemistries; (iii) this tactic is straightforward and unambiguous, meaning that researchers are expected to arrive at identical TOF values for a given data set; and, finally, (iv) the TOFs are under- rather than over-estimated by this approach, which is preferred to reporting artificially inflated performance after accounting for accessible “active” catalytic sites, for example. At 5.0 mol % catalyst (relative to substrate), the catalytic activity of these aged AgNCs is high enough ($k_{\text{app}} = 0.137 \text{ s}^{-1}$, $\text{TOF} = 1700 \text{ h}^{-1}$) to warrant a follow-up study as a laboratory supplement, with a possible focus being the elucidation of changes in catalytic activity as the colloids age with concomitant LSPR band shifting between yellow and blue colloids; however, such a study is not the primary focus of this visual demonstration and will instead be the topic of a more thorough investigation at a later date. We also performed this reduction using PVP-stabilized yellow and blue colloidal AgNCs as catalysts, but the observed activity was inhibited by the strong, bulky PVP ligand (Figure S3).

Finally, the production of these colloidal AgNCs was scaled from 5.0 to 100.0 mL with little to no observed changes in the rate of color progression. We do, however, recommend that such scaling be performed in a way which makes the colors easily visible to the audience, as increasing the path length of the solution can mute some of the transitions in poor lighting conditions. For this reason, smaller volumes are preferred in conjunction with a projector system to display the reaction to larger audiences.

CONCLUSION

In this demonstration, we present the simple, solvent-based synthesis of a yellow colloid of AgNCs, which slowly aggregates, exhibiting a vibrant color transition to orange, red, purple, and finally blue over a 3 min period. The readily available reagents and minimal preparation time make this a very accessible visual aid for undergraduate and advanced high school chemistry courses to discuss the nanoscale effects responsible for the macroscale observations of colloids. Further, the slow and vibrant color change is in stark contrast to the much more rapid transitions observed for similar syntheses, expanding the time frame for concurrent discussion. We additionally show that PVP can be added to the colloid to arrest the color progression, providing a route toward long-term storage or side-by-side comparison of target colors. We propose a follow-up laboratory experiment wherein these AgNCs are used at different points in their aggregation as catalysts in the reduction of nitroarenes to teach the effects of colloid dispersion on catalytic activity, an important lesson for new nanochemists to learn.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available on the ACS Publications website at DOI: 10.1021/acs.jchemed.9b00638.

Additional information about the AgNC synthesis procedure along with further UV–vis and TEM characterization results (PDF, DOCX)

AgNP demo in 96% EtOH/4% H_2O (MP4)

AgNP demo in 100% H_2O (MP4)

AgNP demo in 100% EtOH (MP4)

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Notes

The authors declare no competing financial interest.

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